

Doom PWAD QA Report

Run 1acda2c1-05d2-431f-867e-02ab2e635f6c | Map E1M1 | Outcome pwad_crash

Executive Summary

This is an autonomous QA test of map E1M1 using a lockstep LLM/MCP test harness. The test was designed to evaluate map stability, navigation, and resource balance; however, the run resulted in a 'pwad_crash' outcome, preventing any gameplay execution.

Pass/fail:

map_navigation: FAIL

combat_engagement: LIMITED

resource_balance: LIMITED

secret_coverage: LIMITED

overall_verdict: FAIL

- The map did not initialize into a playable episode.
- No spawned combat state was available.
- Resource balance cannot be evaluated before runtime initialization.
- Secret coverage cannot be evaluated before runtime initialization.

The map failed to initialize under the ViZDoom environment, with the preflight check timing out after 15 seconds. No gameplay data was generated, and no LLM calls were made as the runtime never reached a playable state.

Test Item And Environment

The test targeted map E1M1 using the Freedoom1 IWAD at difficulty level 1. Static analysis indicates a map size of 2048x2560 units, containing 102 enemies, 138 items, 3 keys, 16 weapons, and 7 secret sectors. The map exhibits high static complexity with a total of 420 sectors and 2742 linedefs.

The environment utilized ViZDoom as the game engine, governed by a lockstep LLM/MCP interface. The test configuration specified the gemini-3.1-flash-lite model and a 600-tick cap.

Metric	Value	Metric	Value
Status	failed	Duration	None
Actions	0	LLM calls	0
Final HP	None	Kills	None
Position samples	0	Movement units	0.0
Recording	Not produced		

Static Analysis Context

Metric	Raw WAD	Difficulty 1	Notes
Enemies	102	63	39 hidden by skill/multiplayer flags
Items	138	133	5 hidden by skill/multiplayer flags
Secrets	7	7	Sector special 9
Geometry	2742 linedefs	420 sectors	2048 x 2560 units

Selected-difficulty estimate: hard. Health ratio: 0.0952. Ammo ratio: 0.1207.

Coverage Evaluation

Coverage was limited to static analysis only. Because the map failed to load in the ViZDoom engine, no sector exploration, combat engagement, or item collection occurred. No meaningful gameplay QA evaluation is possible for this run.

Objectives Covered

- **Runtime failure capture:** Map E1M1 could not be loaded safely by ViZDoom (preflight timed out after 15s).
- **Report artifact:** A crash report JSON/PDF was generated.
- **Defect payload:** 2 defect(s) attached to the run.

Objectives Omitted

- **Navigate the entire map:** Map could not be initialized by the test runtime
- **Engage all enemy types encountered:** Map could not be initialized by the test runtime
- **Find secrets where accessible:** Map could not be initialized by the test runtime
- **Stress-test geometry, doors, lifts, and switches:** Map could not be initialized by the test runtime
- **Document resource balance issues:** Map could not be initialized by the test runtime

Uncovered attributes: All gameplay-related attributes, including scripting, geometry traversal, and enemy encounters, were uncovered due to the initialization failure.

Defects And Risks

A total of two defects were identified. The most impactful is a critical PWAD initialization failure preventing all testing; additionally, a minor discrepancy exists regarding difficulty-specific spawn flags.

Patterns: Insufficient defects to identify patterns.

Severity	Priority	Type	Tick	Description
1	1	pwad_crash	0	PWAD crashed or failed to initialize PWAD crashed or failed to initialize under the configured ViZDoom/ Freedoom test environment.
3	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 1, 39 of 102 enemies and 5 of 138 items from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.

Risk Areas

- **Map Initialization:** high The map fails to load in the test engine, preventing any automated verification.

Activity Log Summary

1) Static analysis phase (successful); 2) Map exploration phase (failed/aborted); 3) Combat phase (failed/aborted); 4) Defect detection phase (completed); 5) Report generation phase (completed). Gameplay phases could not be timed as the engine did not initialize.

The run failed during the startup stage, preventing all subsequent gameplay activities.

Limitations And Recommendations

The automated playthrough did not test any map features, including doors, teleporters, or traps, due to the initialization failure.

Good Quality Areas

- **Static Analysis Data:** The pre-run data extraction successfully identified map statistics despite the runtime crash.